Luck Is All You Need @ IEEE CS CUET ML Contest 2.0 – Is an LLM Enough? A Retrieval-Reader-Selector Pipeline for Bangla Factual Question Answering

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***Abstract*—Bangla factual question answering over a large knowledge base requires both strong evidence retrieval and precise answer extraction. In the Intra CUET ML Contest 2.0 task, titled "Are You Sure LLM Is Enough?", the system had to answer Bangla factual questions using a provided knowledge base and structured train-test files. The competition dataset was created using resources from the AI4Bharat Indic-Rag-Suite dataset, while the subset creation and knowledge base construction process was released by the organizers through a Kaggle notebook. We properly acknowledge and cite both resources in this report. This paper presents the final system of Team *Luck Is All You Need*, which combines hybrid retrieval, evidence-window construction, BanglaBERT-based reranking, mDeBERTa extractive reader fine-tuning, candidate voting, and controlled Qwen2.5-7B-Instruct answer selection. Instead of using an LLM as a free generator, our final method uses the LLM only as a selector over reader-generated candidate answers. The final submission, `submission_b3_direct_qwen_medium.csv`, achieved a public leaderboard score of 0.61950 and a private leaderboard score of 0.64512.**

***Index Terms*—Bangla question answering, retrieval-augmented generation, hybrid retrieval, BM25, dense retrieval, reranking, extractive QA, mDeBERTa, BanglaBERT, Qwen**

I. Introduction

Large language models (LLMs) have shown strong performance in many natural language processing tasks. However, factual question answering over a fixed knowledge base still requires careful grounding. In such settings, the model must not only generate fluent text but also locate the correct evidence and return a short exact answer. This is especially challenging for Bangla factual QA because the expected answer is often a short phrase, date, title, location, or named entity, while the supporting evidence may appear inside long and noisy Bengali Wikipedia-style passages.

The competition task, "Are You Sure LLM Is Enough? – Intra CUET ML Contest 2.0", required teams to answer Bangla factual questions using the provided dataset and knowledge base. The competition dataset was created using resources from AI4Bharat Indic-Rag-Suite [1], [2]. The subset creation and knowledge base construction pipeline used for the contest was provided by the organizers in a Kaggle notebook [3]. Indic-Rag-Suite is a resource for Indic-language retrieval-augmented generation and contains question-answer-passage style resources for Indian languages, making it a relevant foundation for this Bangla QA competition.

A naive LLM-only system can produce unsupported or overly long answers. A retrieval-only system can retrieve related passages but still fail to extract the exact answer. Therefore, our final solution follows a staged retrieval-reader-selector design. The system first retrieves candidate evidence, then reranks and converts it into focused windows, then extracts candidate spans using a fine-tuned reader, and finally applies a controlled LLM-based selector to choose the final answer.

The main contributions of our work are:

- A hybrid retrieval system combining BM25 sparse retrieval and BGE-M3 dense retrieval for high-recall evidence search.
- A focused evidence-window construction strategy to reduce long-context noise.
- A BanglaBERT cross-encoder reranker trained with hard negatives to improve evidence precision.
- A fine-tuned mDeBERTa extractive reader for candidate answer span generation.
- A controlled Qwen2.5-7B-Instruct answer selector that chooses from existing candidate answers instead of freely generating unsupported text.

II. Dataset and Task Description

The provided data consisted of training questions with answers, test questions, a sample submission file, and a large Bangla knowledge base. The training set contained 3,000 question-answer examples, while the test set contained 1,500 questions. During our processing, the knowledge base was observed to contain approximately 56.5 million characters and around 6,993 article-like units.

The dataset was created using resources from AI4Bharat Indic-Rag-Suite [1], [2]. The contest organizers also provided the subset creation and knowledge base construction pipeline through a Kaggle notebook [3]. We acknowledge both of these sources as the foundation of the competition dataset.

The task output was a short textual answer for each test question. Since the official score rewarded answer overlap, the system had to avoid both missing key tokens and producing unnecessarily long responses. The major difficulty was the gap between long retrieved contexts and short expected answers. This motivated our final pipeline, where retrieval recall, evidence precision, and answer selection were optimized separately.

III. Methodology

Fig. 1 shows the overall final methodology. The system is intentionally modular: each stage reduces one type of error before passing information to the next stage.

A. Hybrid Retrieval

The first stage builds a high-recall retrieval system over the knowledge base. We used BM25 for sparse lexical retrieval and BGE-M3 embeddings with FAISS for dense semantic retrieval. BM25 is useful when important words from the question directly match the evidence text, while dense retrieval helps when the wording differs but the meaning remains similar. The retrieved results from both systems were combined to create a broad candidate set.

The goal of this stage was not to immediately return an answer. Instead, it ensured that the answer-containing evidence was likely to appear among the top candidate chunks. This became the foundation for later reranking and reading.

B. Evidence Window Construction

Retrieved chunks were often too long and contained multiple facts. Passing such long contexts directly to an extractive reader can confuse the model, especially when several similar entities or dates appear in the same passage. To solve this problem, we converted retrieved chunks into shorter evidence windows.

Several window types were used, including keyphrase-centered windows, title-aware windows, sentence-based windows, and overlapping context windows. For training, answer-positive windows were created when the gold answer appeared in the context. This made the reader training and inference settings more consistent.

C. BanglaBERT Reranking

After window generation, we trained a BanglaBERT-based cross-encoder reranker. The reranker received a question and an evidence window, then predicted how useful the window was for answering the question. Hard negatives were important in this stage. These were windows that looked relevant according to retrieval but did not contain the correct answer. Training with such examples helped the reranker distinguish truly answer-supporting evidence from merely topic-related passages.

The top reranked windows were passed to the extractive reader. This reduced noise and improved the quality of candidate answer spans.

D. mDeBERTa Extractive Reader

The reader stage used a fresh mDeBERTa model fine-tuned as an extractive QA model. The model learned to predict start and end positions of the answer span inside an evidence window. During inference, the reader generated multiple answer candidates from the best reranked windows.

Instead of trusting only the highest-probability candidate, we used a scoring and voting mechanism. Candidate answers were compared based on reader confidence, reranker score, frequency across windows, answer length, and source diversity. This produced the strong B3 direct answer set.

E. Controlled Qwen Answer Selection

The final step used Qwen2.5-7B-Instruct as a controlled selector. Importantly, Qwen was not used as a free-form answer generator. It received the question, selected evidence, B3 reader outputs, and candidate answers, and then selected the most appropriate answer from the candidate set. The output was accepted only if it matched an existing candidate answer.

This design allowed us to use the reasoning ability of an instruction-tuned LLM while avoiding unsupported hallucination. The final medium-selection strategy produced our best submission.

IV. Experimentation

We improved the system over several stages. Early approaches were useful for diagnosing the bottleneck, while the final approach combined the strongest components. Table I summarizes the main experiment progression.

A. Earlier Attempts

The Sprint 2 baseline showed that hybrid retrieval was useful, but retrieval alone could not ensure exact answer extraction. In Sprint 3, an initial reasoning-field-based reader fine-tuning attempt was explored. This was later discarded because the reasoning-derived contexts did not match the actual KB evidence used during inference. The recovery system improved performance by fine-tuning on answer-positive KB chunks.

Later, short-window and gated-reader submissions improved answer conciseness. The largest jump came from the B3 system, where BanglaBERT reranking and a fresh mDeBERTa reader were combined. The final Qwen selector then added a smaller but important improvement by resolving uncertain candidate choices.

V. Results and Analysis

Table II presents the final comparison between the strongest reader-only submission and the final Qwen-selected submission.

The B3 direct system already produced a strong private score of 0.63588. Adding Qwen-based controlled selection increased the private score to 0.64512. The improvement suggests that the LLM selector was useful mainly in uncertain cases where multiple candidate spans were plausible.

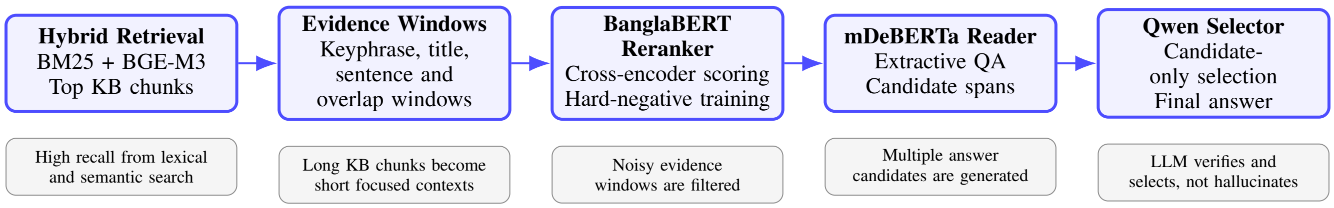

Fig. 1. Final methodology of Team *Luck Is All You Need*: hybrid retrieval, evidence-window construction, reranking, extractive reading, and controlled LLM-based answer selection.

TABLE I
EXPERIMENT PROGRESSION AND LEADERBOARD SCORES.

Stage	Main Idea	Public	Private
Sprint 2 Baseline	Hybrid retrieval with basic answer generation. This established the first retrieval infrastructure but answer extraction was still weak.	0.52723	0.51452
Sprint 3 Recovery	KB-positive reader fine-tuning and answer selection after an earlier reasoning-based reader attempt failed to generalize.	0.57626	0.58104
Sprint 3 Improved	Fast short-window gated reader with stronger filtering and concise answer preference.	0.58676	0.57976
Sprint 3 Improved-2	Fresh mDeBERTa reader trained on focused windows.	0.59474	0.58468
B3 Direct	Reranked evidence windows with fresh mDeBERTa reader and candidate voting.	0.61634	0.63588
Final Qwen Medium	B3 reader candidates with controlled Qwen answer selection.	**0.61950**	**0.64512**

TABLE II
BEST FINAL RESULTS.

Submission	Public	Private
`submission_b3_direct.csv`	0.61634	0.63588
`submission_b3_direct_qwen_medium.csv`	**0.61950**	**0.64512**

The key observation is that the best system was not an LLM-only solution. Retrieval determined whether the answer evidence was available. Reranking determined whether the reader saw the best evidence. The extractive reader generated grounded candidate answers. Finally, the LLM selector improved final choice quality while remaining constrained to candidate spans.

VI. LIMITATIONS

The system still depends on retrieval quality. If the correct passage is not retrieved or is removed during reranking, the reader and selector cannot recover the answer. The Qwen selector is intentionally constrained to candidate answers, which reduces hallucination but prevents it from correcting cases where the right answer is missing from all candidates. The full pipeline is also computationally expensive because it requires dense retrieval, reranking, reader inference, and LLM selection.

VII. ETHICS AND DATASET ACKNOWLEDGMENT

We acknowledge that the competition dataset was created using resources from AI4Bharat Indic-Rag-Suite [1], [2]. We also acknowledge the organizer-provided subset creation process and knowledge base construction pipeline released through Kaggle [3]. All submitted systems were developed using the competition-provided data and allowed resources. No test labels were used during development.

Because factual QA systems can produce incorrect answers if retrieval fails, such systems should be used with proper evidence checking in real applications. Our design reduces hallucination by grounding answers in retrieved evidence and restricting the LLM to candidate selection.

VIII. CONCLUSION

This paper presented the final solution of Team *Luck Is All You Need* for “Are You Sure LLM Is Enough? – Intra CUET ML Contest 2.0”. The final system combined hybrid retrieval, focused evidence windows, BanglaBERT reranking, mDeBERTa extractive reading, candidate voting, and controlled Qwen-based answer selection. The best submission achieved 0.61950 public and 0.64512 private leaderboard scores. The main lesson from the competition was that an LLM alone was not enough; the strongest result came from combining retrieval, reranking, span extraction, and controlled LLM-based selection.

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